AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 12 · STUDENT LESSON

Nearest Neighbors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 22: nearest ties need a predefined policy.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Compute absolute distances and rank training neighbours. Try the recall before opening notes.

Your learning target

I can compute absolute **distance**s and rank **training neighbour**s.

I can apply a declared **neighbour** count and **distance** tie rule.

I can evaluate **neighbour prediction**s without using validation labels as **neighbour**s.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.



Your target

LEARNING TARGET **Training** lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

Builder name and experiment date:

Readiness for the connected explanation

What does a fitted depth-one tree store, and what does it do with a new example?

This week's end goal

By the end, I can explain nearest **neighbor**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Nearest Neighbors case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use nearest **neighbor**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

ID	Reading in cm	Label
A	2	blocked
B	4	clear
C	8	clear
D	10	blocked
E	12	clear

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

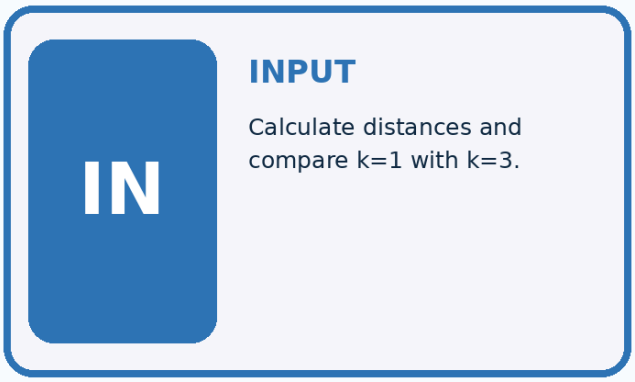
See how it works

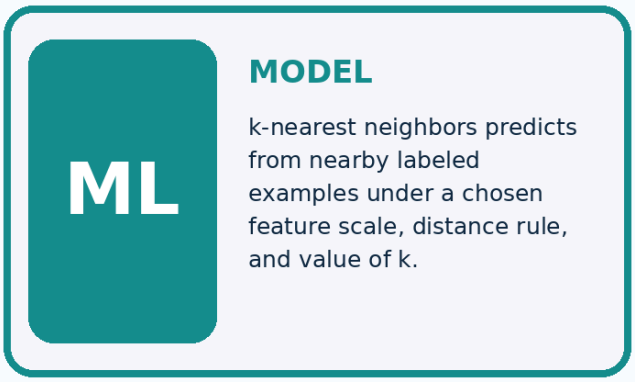
> Follow the mechanism for Nearest Neighbors. Connect each labelled input, deciding step and result.

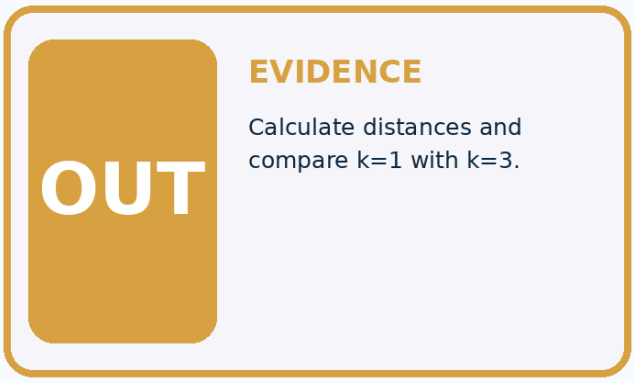
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Calculate **distance**s and compare k=1 with k=3.

MODEL IDEA k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD New leaf has normalized features (2, 3). Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
query = 3
ranked = sorted((abs(x - query), item_id, label)
                for item_id, x, label in train)
near = ranked[:3]
```

```
labels = [row[2] for row in near]
prediction = "clear"
if labels.count("blocked") > labels.count("clear"):
    prediction = "blocked"
print([row[1] for row in near])
print(prediction)
```

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

If k were 1 under the same **distance** tie rule, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

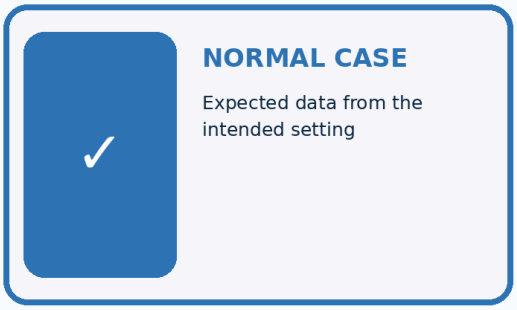
Compare the cases

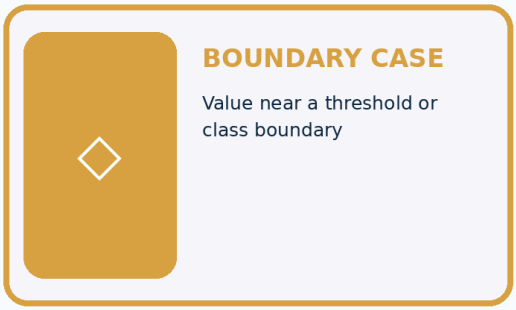
> Compare the supplied cases for Nearest Neighbors. Identify the detail that changes the conclusion.

EVALUATION DESIGN

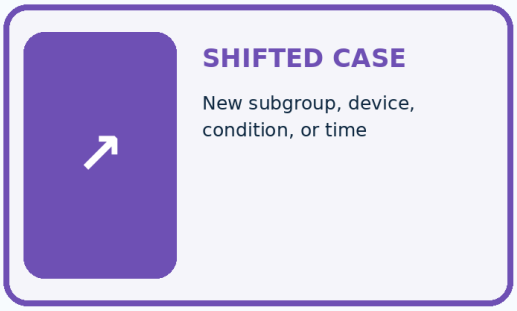
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Move the query to length7 cm.

Predicted result

Prediction: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Why the result follows

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

STUDENT LESSON · 5 OF 18

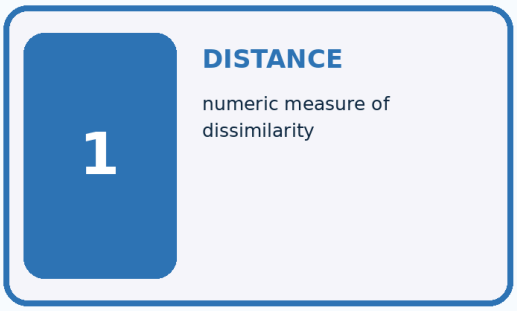
Words that unlock the idea

> Use the vocabulary of Nearest Neighbors to explain a concrete decision, rather than only memorising definitions.

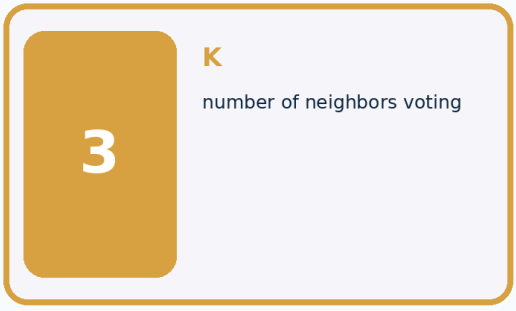
VOCABULARY

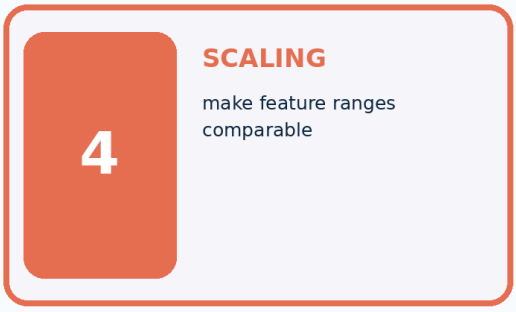
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
distance	numeric measure of dissimilarity	
neighbor	nearby labeled training example	
k	number of neighbors voting	
scaling	make feature ranges comparable	

MISCONCEPTION TO REPAIR Nearest **neighbor** means the photo that looks most similar to a person.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Distance	A numerical measure of difference between selected feature values
Neighbour	A stored training example close under the chosen distance
k	The number of neighbours selected for a prediction
Vote	A contribution of a selected neighbour's label to the predicted class

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Nearest Neighbors, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 12: Nearest Neighbors

STEP	EVIDENCE AND REASONING
1	New leaf has normalized features (2, 3).
2	Using Manhattan distance, add the absolute coordinate differences. Training leaves at (2, 2), (4, 3), and (1, 5) have distances 1, 2, and 3 from (2, 3).
3	With k=1, nearest label wins; with k=3, majority among all three wins.
4	Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Step 1

STEP 1 New leaf has normalized features (2, 3).

Step 2

STEP 2 Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

Step 3

STEP 3 With k=1, nearest label wins; with k=3, majority among all three wins.

Step 4

STEP 4 Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Worked explanation · Nearest Neighbors

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Nearest Neighbors

Let a nearest-**neighbour** picture show each **distance** before voting.

The facts we will use

Supplied facts

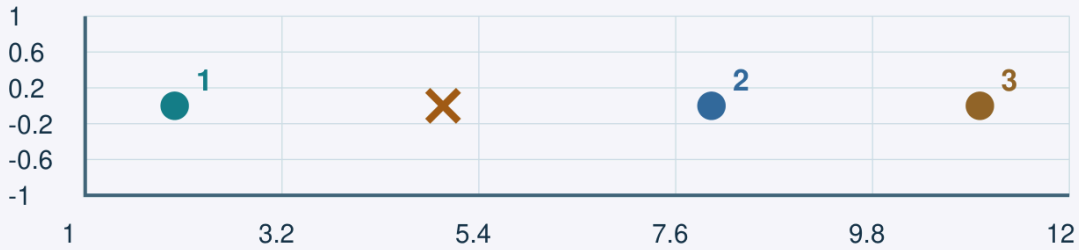
Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm. **Distance** is absolute difference; ties use alphabetically earlier ID.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/12

Nearest Neighbors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Measure distances

To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.

2. Apply k=1 rule

A and B tie nearest; declared ID tie-break chooses A, predicting healthy.

3. Compare k=3

All three vote: two spotted and one healthy, so k=3 predicts spotted.

CHANGE ONE THING

Move the query to length7 cm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.
2. Apply k=1 rule	A and B tie nearest; declared ID tie-break chooses A, predicting healthy.
3. Compare k=3	All three vote: two spotted and one healthy, so k=3 predicts spotted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Length alone is a toy feature, not a disease diagnosis. **Distance** choices and feature **scaling** can change **neighbour**s.

Keep the stated limit

Read the labelled picture: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Move the query to length7 cm. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Nearest Neighbors claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Nearest neighbor means the photo that looks most similar to a person. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Nearest **neighbor** means the photo that looks most similar to a person.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

All **distance**s here use one feature measured in the same unit. With multiple features, their units and scales affect which examples appear close. A large numerical range can dominate a **distance** calculation.

Next week we will address **scaling** and learn any **scaling** values from **training** data only.

Check the claim against the facts

Identical readings from different trips can have different labels, and an exact match does not certify truth. Keep related examples grouped across splits as in Week 6.

Testing an example against a pool containing itself can make a one-**neighbour** score misleadingly high; it does not show performance on a new trip.

A learner adds each labelled validation query to the stored pool, then uses k=1 and reports perfect predictions. Explain what the result fails to test and how to restore the intended comparison.

STUDENT LESSON · 8 OF 18

Practise together

Practise Nearest Neighbors with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Keep k=3 and the original five **training** rows. Validation queries with reference labels are (3,blocked), (9,blocked), (11,clear). Their selected **neighbour** IDs are A B C, C D E, and D E C respectively. Each group **vote**s clear.

Only the last **prediction** matches, so accuracy is 1/3, about 33.3 percent.

The **training**-majority baseline also predicts clear because three **training** labels are clear and two are blocked. It gets the same 1/3 on these validation rows. Record all **distance**s, selected IDs and errors.

Trying more k values is a development choice to document; preserve a separate final test for a final evaluation.

Practise with less help: Apply neighbour and tie rules

Use workbook W2 for the connected example "Nearest **neighbour**s and **distance**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: "A negative difference is a negative **distance**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

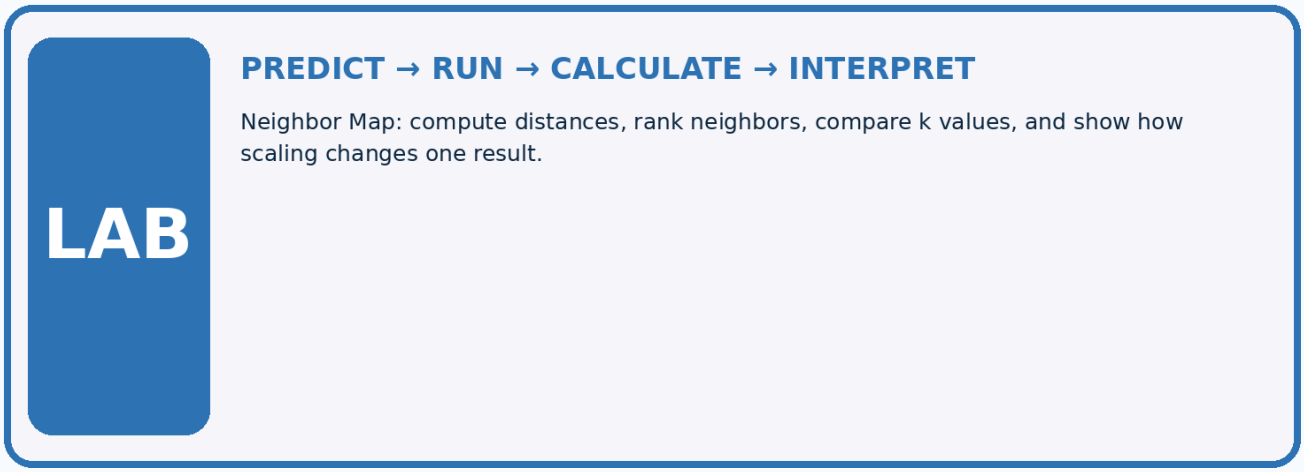
Run the investigation

> Investigate Nearest Neighbors. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. **Neighbor** Map: compute **distance**s, rank **neighbor**s, compare k values, and show how

scaling changes one result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Boundary
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Missing
Prediction: _____
Actual: _____
Calculation / evidence: _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

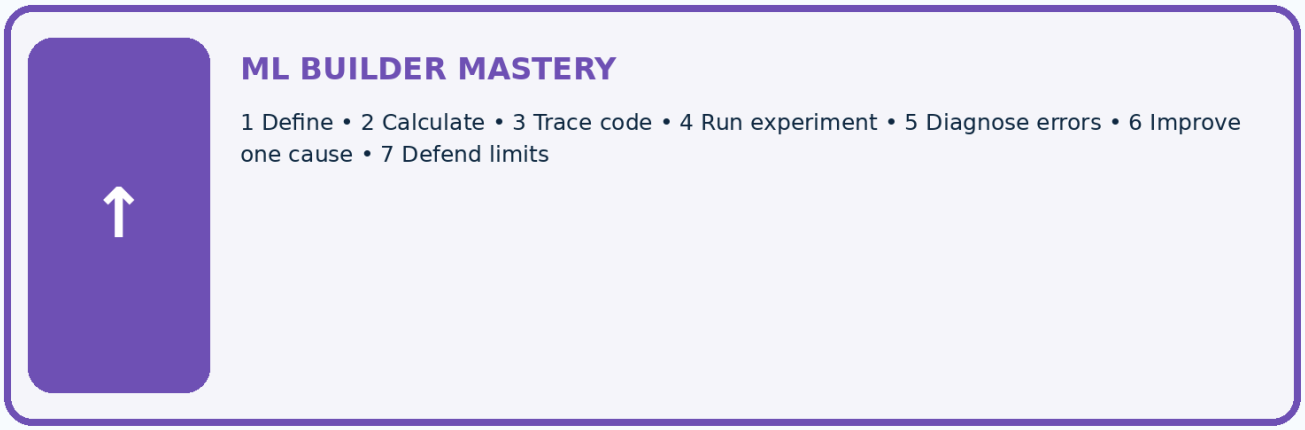
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Nearest Neighbors. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **distance**, **neighbor**, k, **scaling**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Nearest **neighbor** means the photo that looks most similar to a person.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

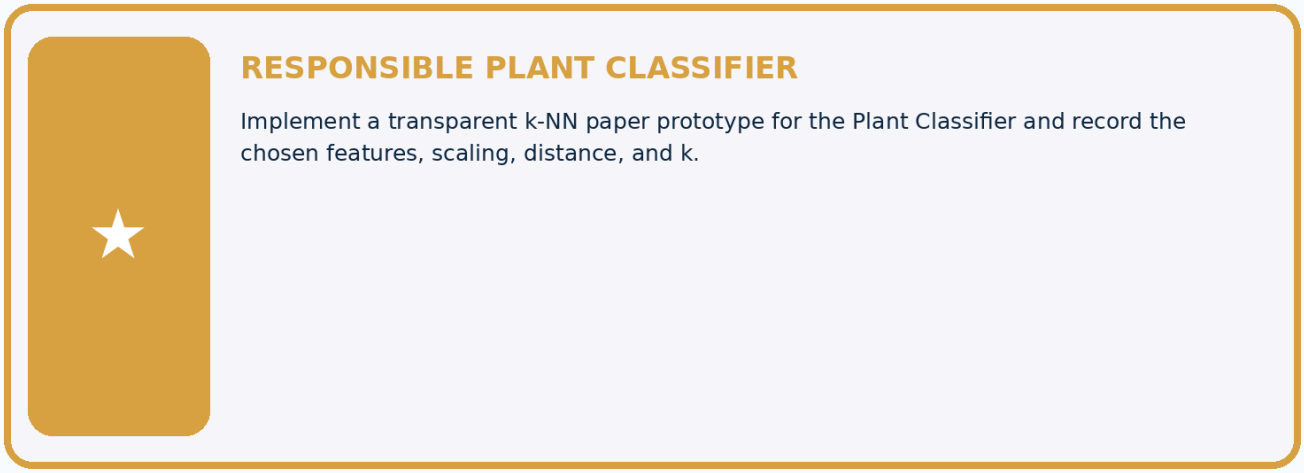
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Calculate **distance**s and compare k=1 with k=3.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Train a Classifier.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Nearest Neighbors case for supported practice. Repeated examples do not become fresh assessments.

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

The question

Find k=1 **neighbour**s and handle ties.

Worked explanation

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Absolute distance from query 6

Evidence or stage	Value or result
Point 2, healthy	4
Point 4, healthy	2
Point 8, spotted	2

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

Find k=1 **neighbour**s and handle ties.

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Training points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute **distance**; for a nearest-**neighbor** tie choose the alphabetically earlier ID. Compare k=1 and k=3.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Training lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Nearest Neighbors, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Let a nearest-**neighbour** picture show each **distance** before voting.

By the end, I can	How I will show it
Compute absolute distances and rank training neighbours	Explain the deciding step, show a trace or test, and state one limit.
Apply a declared neighbour count and distance tie rule	Explain the deciding step, show a trace or test, and state one limit.
Evaluate neighbour predictions without using validation labels as neighbours	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Nearest Neighbors without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compute absolute **distance**s and rank **training neighbour**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Let a nearest-**neighbour** picture show each **distance** before voting.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

1. Measure **distance**s

2. Compare k=3

3. Apply k=1 rule

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare k=3 All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Read the picture in order.

All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move the query to length7 cm. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Repair a missing building block

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not **vote** away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

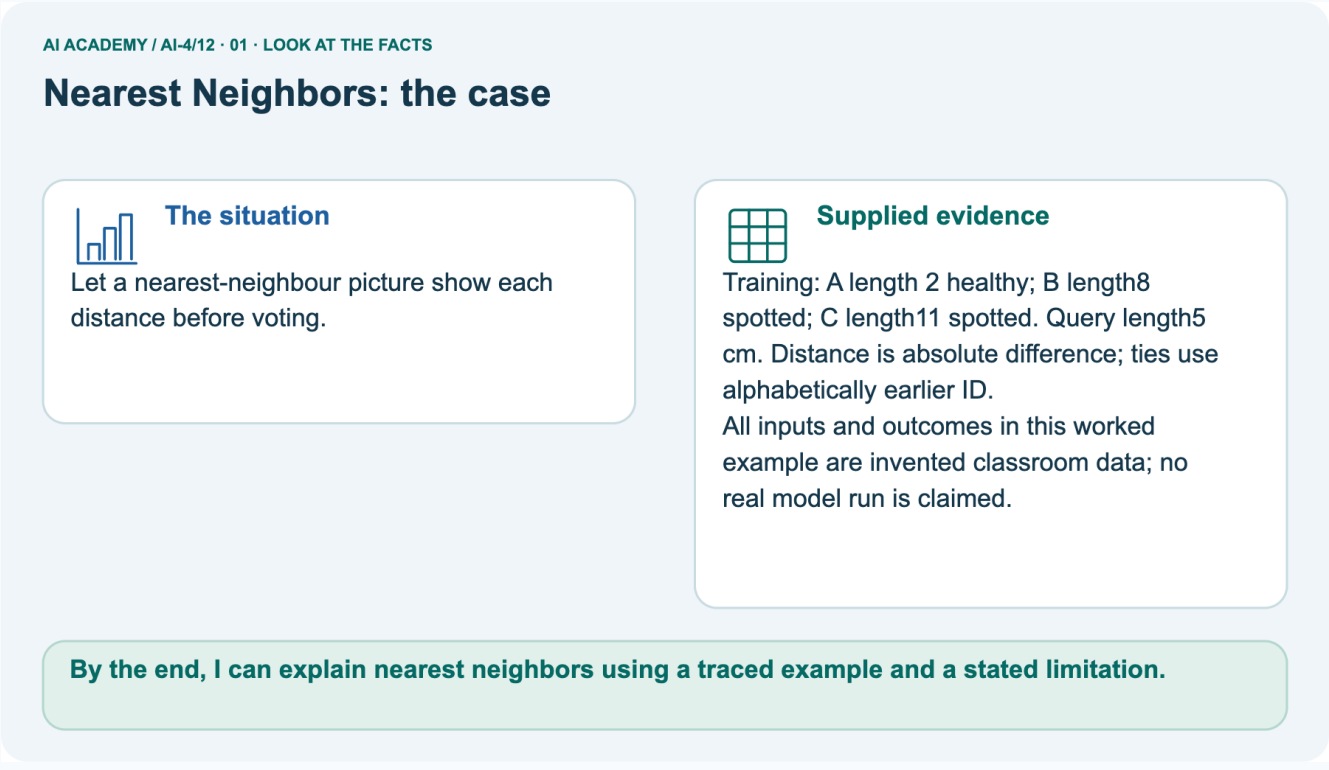
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Nearest Neighbors graphic using its labels, deciding evidence and relevant limitation.

01. Nearest Neighbors: the case



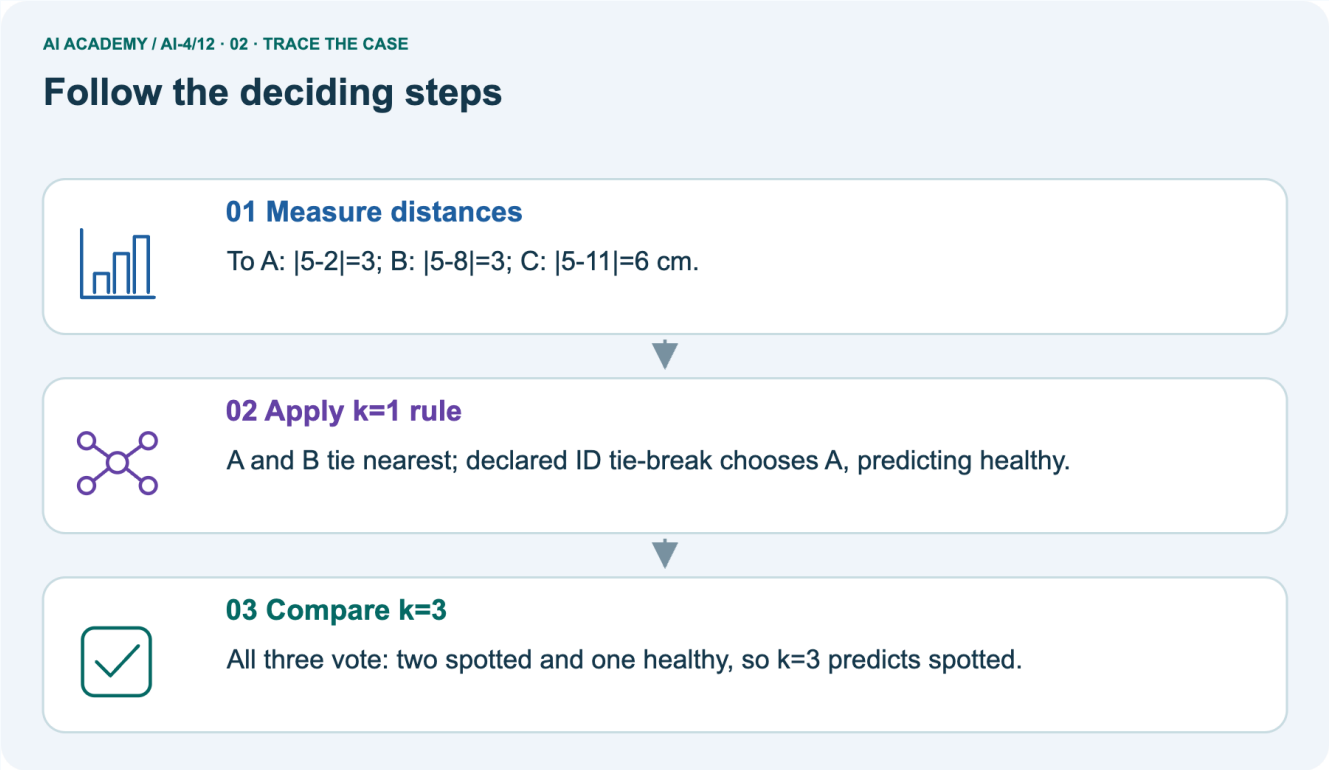
Follow the labelled picture with the matching explanation. Open the vector diagram

Let a nearest-**neighbour** picture show each **distance** before voting. Supplied facts: **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.

Distance is absolute difference; ties use alphabetically earlier ID. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

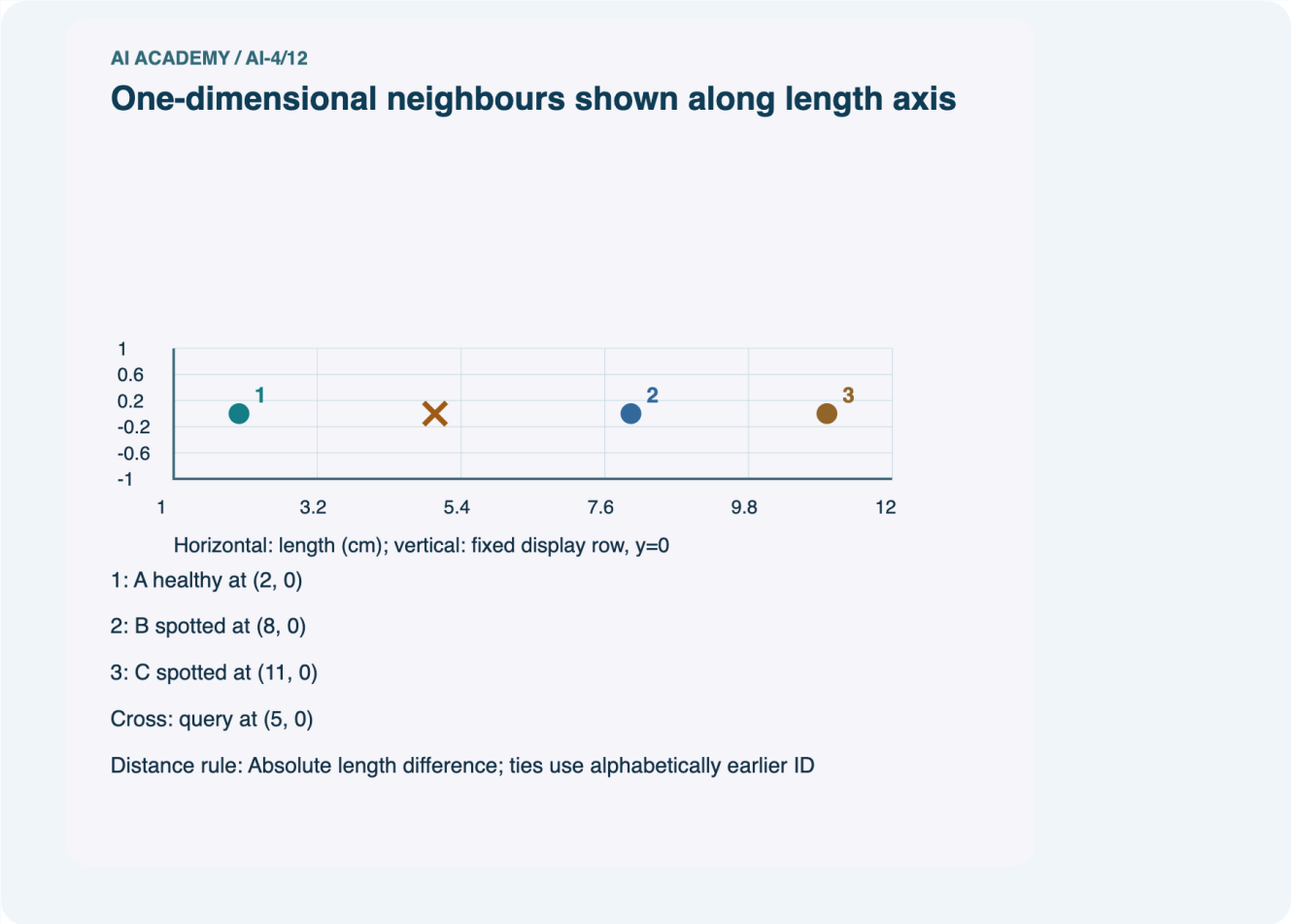


Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm. → Apply k=1 rule: A and B tie nearest; declared ID tie-break chooses A, predicting healthy. → Compare k=3: All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

> Point to each numbered stage and explain which recorded **input** it uses.

03. One-dimensional neighbours shown along length axis



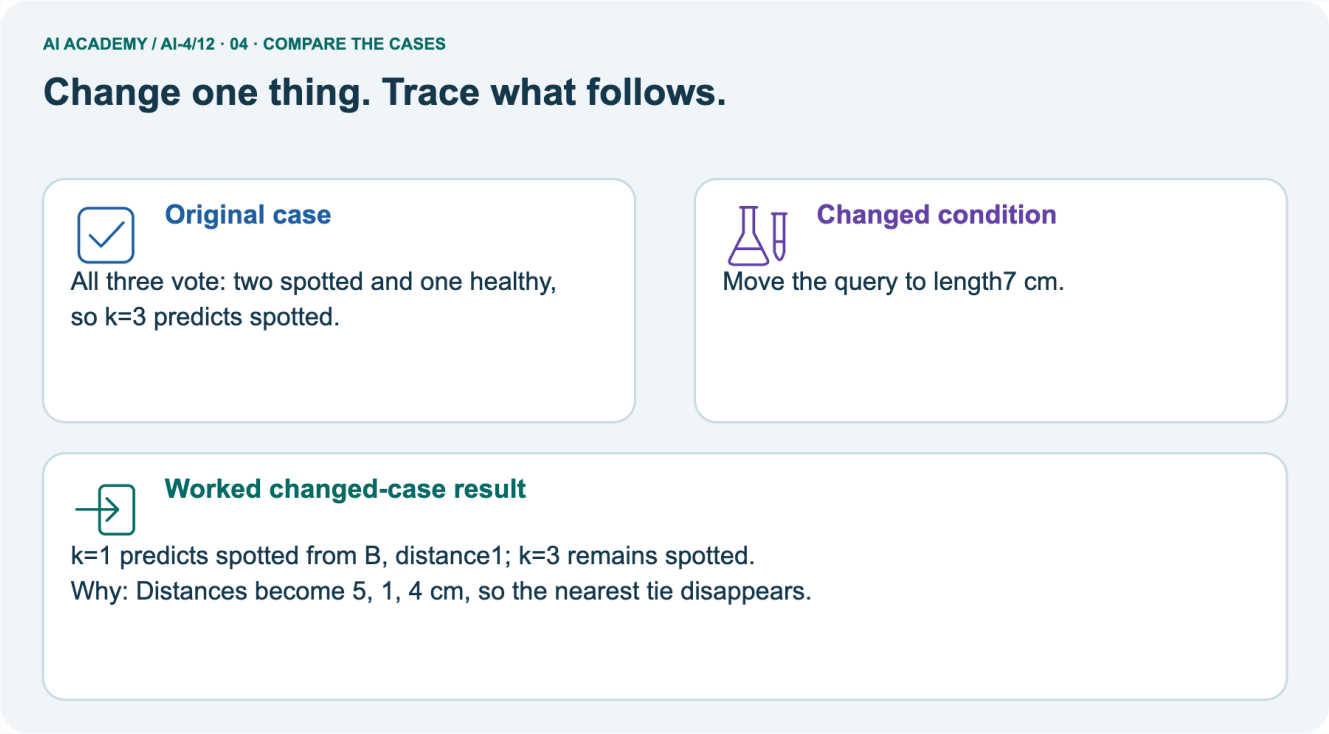
Follow the labelled picture with the matching explanation. Open the vector diagram

One-dimensional **neighbour**s shown along length axis. A healthy: (2, 0). B spotted: (8, 0). C spotted: (11, 0). Query query: (5, 0). Axes: length (cm); fixed display row, y=0.

Distance: Absolute length difference; ties use alphabetically earlier ID.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



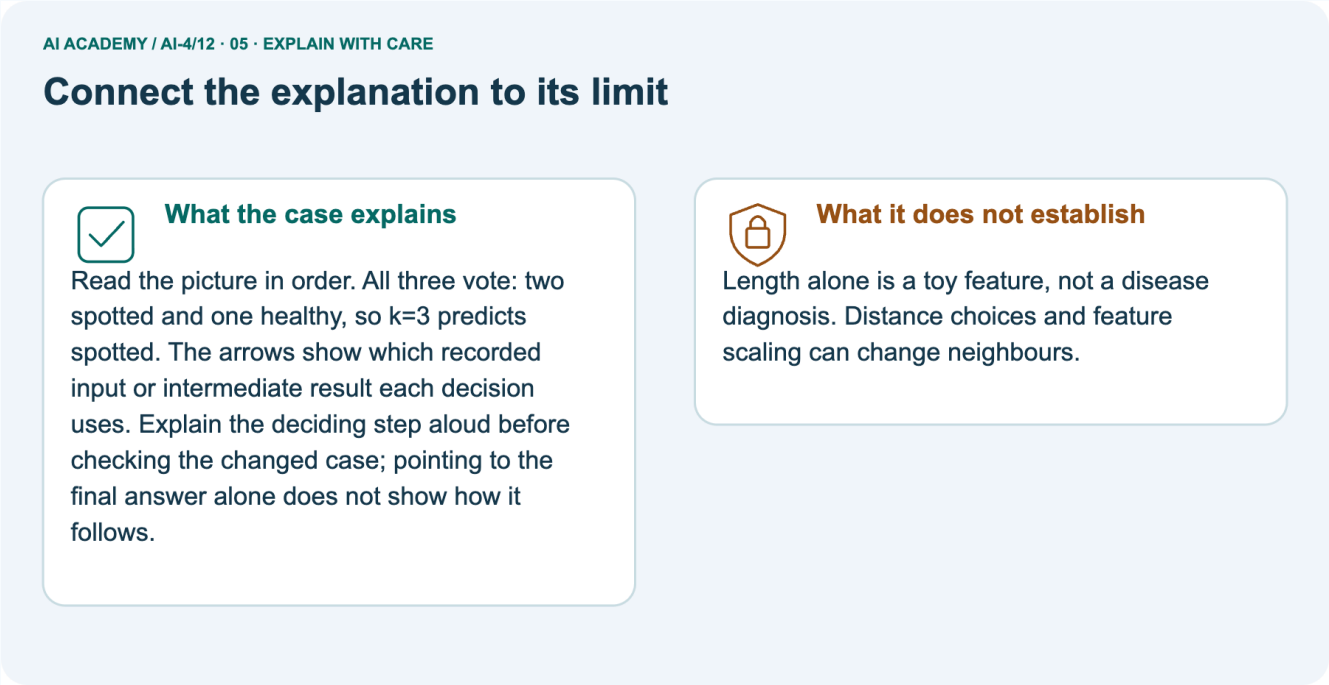
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Changed condition: Move the query to length7 cm. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

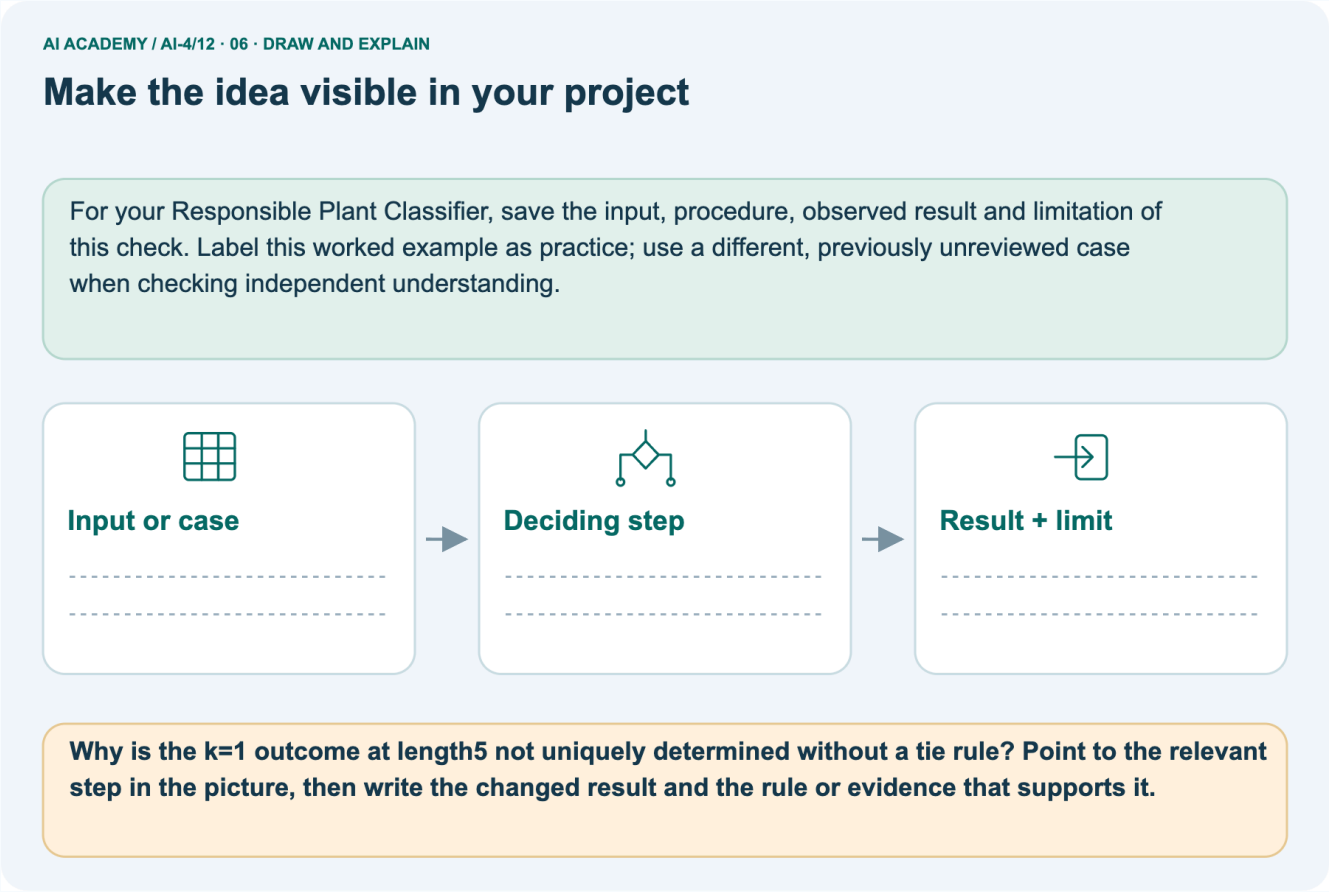
Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Length alone is a toy feature, not a disease diagnosis.

Distance choices and feature **scaling** can change **neighbour**s.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.